

University of London
Imperial College of Science, Technology and Medicine
Department of Physics

I Simply Cannot, and Will Not, Stop Shagging the Kitaev Model

Peru d'Ornellas

Submitted in part fulfilment of the requirements for the degree of
Doctor of Philosophy in Physics of Imperial College of Science,
Technology and Medicine, December 2023

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Contents

1	Introduction	7
1.1	Quantum Hall and Topological Insulators	7
1.1.1	Initial motivation - The quantum hall effect	7
1.1.2	Do some review	7
1.2	Spin Liquids	7
1.2.1	background - anderson RVB, frustrated magnets, X-G Wen and co	7
1.2.2	Kitaev Model	7
1.2.3	Physical materials	7
1.3	Thesis Outline	7
2	Chern Number, Topological Insulators and Local Markers	8
2.1	Abelian Berry Phase and the Chern Number	8
2.1.1	Discrete Case	8
2.1.2	Continuous Case	11
2.1.3	An Example - The Two-Level System	13
2.1.4	Final Notes	16
2.2	The QWZ Model and its Amorphous Counterpart	17
2.2.1	Strip Geometry	19
2.2.2	The Amorphous QWZ Model	22
2.3	Local Markers	22
2.3.1	Kitaev's Local Marker	22
2.3.2	The Bott Index	22
2.3.3	The Chern Marker	22
3	The Crosshair Marker	24
4	Kitaev's Honeycomb Model	25
5	The Amorphous Kitaev Model	26

6	The Star Lattice Kitaev-Heisenberg Model	27
7	Conclusion and Outlook	28

Chapter 1

Introduction

1.1 Quantum Hall and Topological Insulators

1.1.1 Initial motivation - The quantum hall effect

1.1.2 Do some review

1.2 Spin Liquids

1.2.1 background - anderson RVB, frustrated magnets, X-G Wen and co

1.2.2 Kitaev Model

1.2.3 Physical materials

1.3 Thesis Outline

Chapter 2

Chern Number, Topological Insulators and Local Markers

Paragraph introducing the topics covered in this section. Points to hit:

-

2.1 Abelian Berry Phase and the Chern Number

We shall start by introducing the concepts of Berry phase, Berry flux and Chern number, quantities that will be of fundamental importance for understanding the properties of topological insulators.

Berry phase was first introduced in [1], the ideas presented here will largely follow [2], but other resources for understanding these concepts can be found in [3, 4]. These concepts will first be introduced in the discrete case, looking at the Berry phase accumulated as one moves adiabatically around a lattice of quantum states. However in §2.1.2 we will extend the argument to include adiabatic paths around a smooth manifold of states. In §2.1.3 we will give a brief example of such a calculation, demonstrating how these concepts may be applied to periodic quantum systems.

2.1.1 Discrete Case

Suppose we have a quantum system that lives in a Hilbert space $\mathcal{H}$. States in this system are denoted by an equivalence class of vectors in this space. This means that two states that differ only by a complex phase still represent the same state,

$$|\psi\rangle \sim e^{-i\alpha} |\psi\rangle. \quad (2.1.1)$$

Thus we essentially have a degeneracy in our description of the physics taking place. It is possible to make a gauge transformation – shifting the state by some complex phase – without affecting the system in any measurable way.

Now suppose that we have collected a set of states that live in this Hilbert space, $\{|\psi_j\rangle\}$. Rather than acting on each state with the same phase (and thus implementing a global gauge

transformation), you apply a different phase to each state. This amounts to making a local gauge transformation,

$$|\psi_j\rangle \rightarrow e^{-i\alpha_j} |\psi_j\rangle. \quad (2.1.2)$$

The physical meaning of each state must be unaffected, so we expect that any quantity of physical importance should be invariant to our choice of gauge. With this condition in mind we can try to define quantities that compare our states, and test whether their definitions respect gauge invariance. The first quantity we propose is the relative phase between two states.

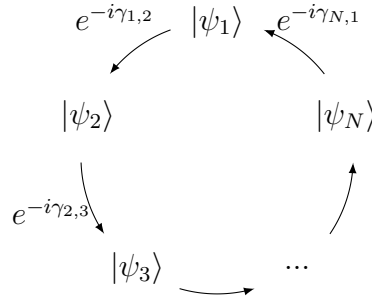
$$e^{-i\gamma_{j,k}} = \frac{\langle \psi_j | \psi_k \rangle}{|\langle \psi_j | \psi_k \rangle|} \quad (2.1.3)$$

Clearly this quantity is invariant under global gauge transformations, but it is definitely not invariant under a local transformation since it changes according to

$$e^{-i\gamma_{j,k}} \rightarrow e^{-i(\gamma_{j,k} + \alpha_j - \alpha_k)}. \quad (2.1.4)$$

Let's try something else.

We pick a set of N (≥ 3) states $|\psi_j\rangle$ and arrange them in a loop.



We can now define a new quantity called the Berry phase:

$$e^{-i\gamma^B} = \frac{\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle}{|\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle|} \quad (2.1.5)$$

Another way of writing this would be

$$\gamma^B = -\arg \text{Tr} [\rho_1 \rho_2 \dots \rho_N], \quad (2.1.6)$$

where trace is over the whole system, and the ρ states are density matrices

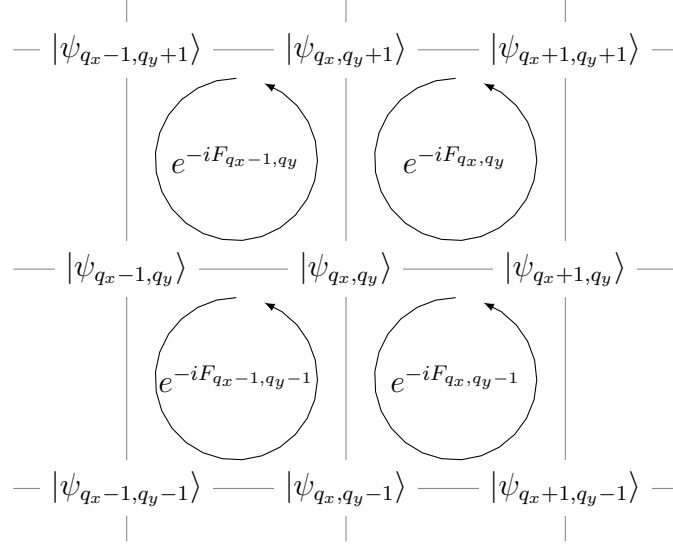
$$\rho_i = |\psi_i\rangle \langle \psi_i|. \quad (2.1.7)$$

This quantity is clearly gauge-invariant, since any phase you attach to a state $|\psi_m\rangle$ will show up twice, once from $\langle \psi_m | \psi_{m+1} \rangle$ and once from $\langle \psi_{m-1} | \psi_m \rangle$. The phase will have opposite sign in each of these expressions so they cancel out. Furthermore, we can see that the Berry phase around a loop is simply the sum of all the relative phases around that loop modulo 2π .

$$\gamma^B = \sum_{n=1}^N \gamma_{n,n+1} \mod 2\pi \quad (2.1.8)$$

Berry Flux

We now imagine that we have some massive amount of states arranged in a grid. These are labeled according a pair of parameters that we will call q_x and q_y , $|\psi_{q_x, q_y}\rangle$. Note that we have used q as a label here because it will turn out that in periodic quantum systems, these states correspond to the Bloch eigenstates, and q will correspond to the $\mathbf{k}$ parameter in our $|u_{\mathbf{k}, \mathbf{n}}\rangle$ states.



We introduce a new quantity called the Berry flux. This is defined to be the Berry phase around a unit loop in our grid,

$$F_{q_x, q_y} = -\arg \text{Tr} [\rho_{q_x, q_y} \rho_{q_x+1, q_y} \rho_{q_x+1, q_y+1} \rho_{q_x, q_y+1}] , \quad (2.1.9)$$

where each ρ represents the density matrix

$$\rho_{q_x, q_y} = |\psi_{q_x, q_y}\rangle \langle \psi_{q_x, q_y}| . \quad (2.1.10)$$

This quantity has a neat property. F_{q_x, q_y} is just the sum (mod 2π) of the relative phases along each edge of the plaquette starting at point (q_x, q_y) . Thus, if we wanted to calculate the Berry phase around some arbitrary loop that contained multiple plaquettes, we could just sum the value of F_{q_x, q_y} for every plaquette contained in that region. Each relative phase flips sign depending on which way round it is evaluated. Any internal edge is shared between two plaquettes and so will appear twice in the sum of Berry fluxes, with opposite signs. This means all internal edges have their contribution to the total phase cancelled out and we are left with the sum of the relative phases only around the boundary of our region.

$$\exp \left[-i \sum_{\text{inside region}} F \right] = \exp \left[-i \sum_{\text{boundary}} \gamma \right] \quad (2.1.11)$$

This vaguely resembles Stokes' law, however there is one subtlety worth mentioning. All we have required is that the complex exponential of each sum is the same. This does **not** mean that the sum of the Berry flux inside a region is equal to the sum of relative phases around the boundary, since they are allowed to differ by an arbitrary multiple of 2π

Chern Number

Now let's assume that our grid of points (q_x, q_y) has periodic boundaries, with $1 \leq q_x \leq N_x$ and $1 \leq q_y \leq N_y$. That is, we have wrapped our space up to make a torus. The product of all the Berry flux phases is now always equal to 1,

$$\prod_{q_x=1}^{N_x} \prod_{q_y=1}^{N_y} e^{-iF_{q_x, q_y}} = 1. \quad (2.1.12)$$

This is because every edge in our grid is now an internal edge, so they all cancel out! Again, as before, this does not mean that the sum of all Berry fluxes in our system vanishes. It means that the sum must always equal some multiple of 2π . This sum gives us the Chern number, which we will call $\mathcal{Q}$,

$$\mathcal{Q} = \frac{1}{2\pi} \sum_{q_x, q_y} F_{q_x, q_y}. \quad (2.1.13)$$

$\mathcal{Q}$ is necessarily restricted to integer values, since a non-integer value would contradict equation 2.1.12.

2.1.2 Continuous Case

Now we repeat the entire derivation, but applied to continuous quantum systems, rather than a discrete lattice of states. In this new system, rather than a grid of discrete states $|\psi_{q_x, q_y}\rangle$ we are working with some smooth parameter space $\mathcal{P}$ of states.

$$|\psi(\mathbf{q})\rangle, \mathbf{q} \in \mathcal{P} \quad (2.1.14)$$

In this context, a local gauge transformation amounts to applying some $\mathbf{q}$ -dependent phase $\alpha(\mathbf{q})$ to all of the states in our system.

$$|\psi(\mathbf{q})\rangle \rightarrow e^{-i\alpha(\mathbf{q})} |\psi(\mathbf{q})\rangle. \quad (2.1.15)$$

Before we continue, let us define what we mean by a *smooth gauge*. This is a choice of gauge for our states that ensures that the $\mathbf{q}$ -derivative of a state, $\nabla_{\mathbf{q}} |\psi(\mathbf{q})\rangle$ is always well-defined. In general it is not always possible to pick a gauge that is globally smooth over all of $\mathcal{P}$, however it *is* always possible to pick a locally smooth gauge around the neighbourhood of some point in $\mathcal{P}$ ¹.

We now want to look at the relative phase between a state at $\mathbf{q}$ and another state that is infinitesimally displaced from it, at $\mathbf{q} + d\mathbf{q}$. This can be expressed as

$$e^{-i\Delta\gamma} = \frac{\langle \psi(\mathbf{q}) | \psi(\mathbf{q} + d\mathbf{q}) \rangle}{|\langle \psi(\mathbf{q}) | \psi(\mathbf{q} + d\mathbf{q}) \rangle|}. \quad (2.1.16)$$

In the limit of small $d\mathbf{q}$ we can express $|\psi(\mathbf{q} + d\mathbf{q})\rangle$ to first order in $d\mathbf{q}$ as

$$|\psi(\mathbf{q} + d\mathbf{q})\rangle = |\psi(\mathbf{q})\rangle + d\mathbf{q} \cdot \nabla_{\mathbf{q}} |\psi(\mathbf{q})\rangle. \quad (2.1.17)$$

¹Do we need a more in depth justification of this statement? Or is it ok just to state this and move on?

Which means that

$$\langle \psi(\mathbf{q}) | \psi(\mathbf{q} + d\mathbf{q}) \rangle = 1 + \langle \psi(\mathbf{q}) | d\mathbf{q} \cdot \nabla_{\mathbf{q}} | \psi(\mathbf{q}) \rangle. \quad (2.1.18)$$

Now, noticing that $\langle \psi(\mathbf{q}) | \nabla_{\mathbf{q}} | \psi(\mathbf{q}) \rangle$ is pure imaginary, it can be shown that

$$\Delta\gamma = i \langle \psi(\mathbf{q}) | \nabla_{\mathbf{q}} | \psi(\mathbf{q}) \rangle \cdot d\mathbf{q} \quad (2.1.19)$$

$$= \mathbf{A} \cdot d\mathbf{q} \quad (2.1.20)$$

The quantity multiplying $d\mathbf{q}$ is the Berry connection $\mathbf{A}(\mathbf{q})$. Like the relative phase before, it is not a gauge-invariant quantity. Under a gauge transformation it changes according to

$$|\psi(\mathbf{q})\rangle \rightarrow e^{-i\alpha(\mathbf{q})} |\psi(\mathbf{q})\rangle, \quad \mathbf{A}(\mathbf{q}) \rightarrow \mathbf{A}(\mathbf{q}) - \nabla_{\mathbf{q}}\alpha(\mathbf{q}) \quad (2.1.21)$$

Essentially $\mathbf{A}(\mathbf{q})$ is telling you how the phase changes as you move away from each point in $\mathcal{P}$.

As you move around some closed path $\mathcal{C}$ in $\mathcal{P}$, the Berry connection allows you to figure out the total phase accumulated over this route, according to

$$\gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{q}. \quad (2.1.22)$$

The total phase around the loop *is* gauge invariant, one can see this intuitively by realising that it is identical to the Berry phase from the discrete case, but in the limit where the discrete steps become arbitrarily close together. However it is also immediately clear from looking at eqn. 2.1.22 that any contribution to $\mathbf{A}$ from a gauge transformation has the form of a total derivative as in eqn. 2.1.21, so will vanish when integrated over a closed loop.

Berry Curvature

We now want to define a quantity that looks like the Berry flux we defined earlier. We start by rewriting eqn. 2.1.22 in the language of differential forms, defining the one-form

$$A = A_a dq^a. \quad (2.1.23)$$

The Berry phase around some infinitesimally small closed path $\mathcal{C}$ can be expressed as

$$\gamma(\mathcal{C}) = \oint_{\mathcal{C}} A. \quad (2.1.24)$$

Now we can use Stokes' theorem to re-express this as a surface integral over some region $\mathcal{R}$ that has the curve $\mathcal{C}$ as its boundary

$$\gamma(\mathcal{C}) = \int_{\mathcal{R}} dA, \quad (2.1.25)$$

where dA is the exterior derivative of A , a quantity that we will call the Berry curvature, B ,

$$B = dA \quad (2.1.26)$$

$$= \frac{\partial A_a}{\partial q^b} dq^b \wedge dq^a. \quad (2.1.27)$$

This calculation has one important caveat. It is only possible to apply Stokes' theorem when A is smooth over the region $\mathcal{R}$. This means that we need to have already picked a smooth gauge for our states at least in the region we are integrating over! As stated before, this is always possible, but only because we have allowed the region $\mathcal{R}$ to be some small subregion of $\mathcal{P}$.

The Berry curvature is a gauge-independent quantity, as can be seen by substituting the gauge-transformed form of A (eqn. 2.1.21) into the definition.

Now, in a two dimensional parameter space, this integral becomes

$$\gamma(\mathcal{C}) = \int_{\mathcal{R}} dq_x dq_y (\partial_x A_y(\mathbf{q}) - \partial_y A_x(\mathbf{q})), \quad (2.1.28)$$

and we can see that the Berry curvature is a scalar

$$B = \partial_x A_y(\mathbf{q}) - \partial_y A_x(\mathbf{q}). \quad (2.1.29)$$

Similarly in three dimensions the Berry curvature is a vector quantity

$$\mathbf{B}(\mathbf{q}) = \nabla_{\mathbf{q}} \times \mathbf{A}(\mathbf{q}). \quad (2.1.30)$$

Chern Number

As in §2.1.1, we make the assertion that the parameter space $\mathcal{P}$ is a closed manifold such as a torus. The Chern number is defined as the integral of the Berry curvature across the whole parameter space

$$\mathcal{Q} = \frac{1}{2\pi} \int_{\mathcal{P}} B. \quad (2.1.31)$$

In a two-dimensional parameter space (q_x, q_y) , this can be expressed as

$$= \frac{1}{2\pi} \int_{\mathcal{P}} [\partial_x \langle \psi(\mathbf{q}) | \partial_y \psi(\mathbf{q}) \rangle - \partial_y \langle \psi(\mathbf{q}) | \partial_x \psi(\mathbf{q}) \rangle] dq_x dq_y \quad (2.1.32)$$

$$= \frac{1}{\pi} \Im \int_{\mathcal{P}} \langle \partial_x \psi(\mathbf{q}) | \partial_y \psi(\mathbf{q}) \rangle dq_x dq_y \quad (2.1.33)$$

The Chern number here is also restricted to only take integer values. This can be seen by realising that the continuous system is equivalent to a lattice system in the limit where the lattice spacing tends to zero, and so the same argument applies. A more careful justification can be found in [5].

2.1.3 An Example - The Two-Level System

We will illustrate these concepts with the example of a two-state system. Our first step will be to parametrise the possible Hamiltonians in such a system using a vector $\mathbf{d}$. We will then solve this Hamiltonian for the positive and negative energy eigenstates. This allows us to look at how the eigenstates change as we smoothly deform the Hamiltonian, exploring the parameter space of $\mathbf{d}$. By examining how the eigenstates of $\hat{H}$ depend on the parametrisation, we can calculate the Berry curvature of the positive and negative energy states. We then connect this with our

understanding of the Bloch states in a periodic quantum system. We discuss how the bulk Hamiltonian of a periodic system corresponds to the embedding of a closed surface in $\mathbf{d}$ -space and describe how to calculate the Chern number of the bands in such a system.

Here the Hilbert space is $\mathbb{C}^2$. This means that any Hamiltonian we construct must be a 2×2 Hermitian matrix. We will parametrise this in terms of a Pauli vector, $\mathbf{d}$,

$$H = \mathbf{d} \cdot \hat{\sigma} = d_i \sigma^i. \quad (2.1.34)$$

where the σ_i are the Pauli matrices.

The set of possible Hamiltonians is thus a manifold in itself, $\{H\} \cong \mathbb{R}^3$, since $\mathbf{d} \in \mathbb{R}^3$. The first thing we will do is to express the vector $\mathbf{d}$ in standard spherical polar coordinates, giving us a Hamiltonian that looks like

$$H = |\mathbf{d}| \begin{pmatrix} \cos(\theta) & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos(\theta) \end{pmatrix} \quad (2.1.35)$$

This matrix has $H^2 = |\mathbf{d}|^2 \mathbb{1}$, so its eigenvalues are $\pm |\mathbf{d}|$. One can show that its eigenvectors are

$$|+\mathbf{d}\rangle = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad |-\mathbf{d}\rangle = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}. \quad (2.1.36)$$

We have thus arrived at a set of states, $|\pm\mathbf{d}\rangle$ that depend on some parameter $\mathbf{d}$. This means that we can apply what we just figured out in the last section. From eqn. 2.1.30 we can see that the Berry connection and Berry curvature of the $|+\rangle$ state is given by

$$\mathbf{A}^\pm(\mathbf{d}) = i \langle \pm\mathbf{d} | \nabla_{\mathbf{d}} | \pm\mathbf{d} \rangle \quad (2.1.37)$$

$$\mathbf{B}^\pm(\mathbf{d}) = \nabla_{\mathbf{d}} \wedge \mathbf{A}^\pm(\mathbf{d}). \quad (2.1.38)$$

Using this equation it is possible to directly calculate the Berry curvature. We will omit this for the sake of brevity, however the full calculation can be found in Appendix ???. We arrive at the following expression for the Berry curvature of the $|+\rangle$ and $|-\rangle$ states:

$$\mathbf{B}^\pm = \pm \frac{\hat{n}_{\mathbf{d}}}{2\mathbf{d}^2} \quad (2.1.39)$$

where $\hat{n}_{\mathbf{d}}$ is a unit vector pointing in the direction of $\mathbf{d}$.

The Berry curvature is equivalent to a monopole field radiating out from the origin in $\mathbf{d}$ -space, in analogy with electrostatics. Now, suppose we wanted to calculate the Berry phase of the $|+\mathbf{d}\rangle$ state around some path in $\mathbf{d}$ -space, we would want to evaluate

$$\gamma^+(C) = \oint_C \mathbf{A}^+(\mathbf{d}) \cdot d\mathbf{d}. \quad (2.1.40)$$

Re-expressing this in terms of the Berry curvature gives us

$$\gamma^+(C) = \int_{\mathcal{R}} \mathbf{B}^+(\mathbf{d}) \cdot d\mathbf{S}, \quad (2.1.41)$$

where the integral is over a region $\mathcal{R}$ that has the curve $\mathcal{C}$ as its boundary. Since the Berry curvature is just a monopole field with charge $\frac{1}{2}$, this integral ends up evaluating half the solid

angle subtended by the curve $\mathcal{C}$.

$$\gamma^+(C) = \frac{1}{2}\Omega_{\mathcal{C}}. \quad (2.1.42)$$

Chern Number

Now we can connect this with our understanding of Bloch states. We will work with a translationally symmetric lattice Hamiltonian in two-dimensional space. As we showed in section ??, it is always possible to diagonalise this Hamiltonian in the plane-wave basis,

$$\hat{H} = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{H}(\mathbf{k}). \quad (2.1.43)$$

If the unit cell contains two sites, then $\hat{H}(\mathbf{k})$ is a 2×2 matrix that depends on $\mathbf{k}$. We can write it in terms of a $\mathbf{d}$ vector that depends on $\mathbf{k}$.

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \hat{\sigma}, \quad (2.1.44)$$

$$\mathbf{k} = (k_x, k_y). \quad (2.1.45)$$

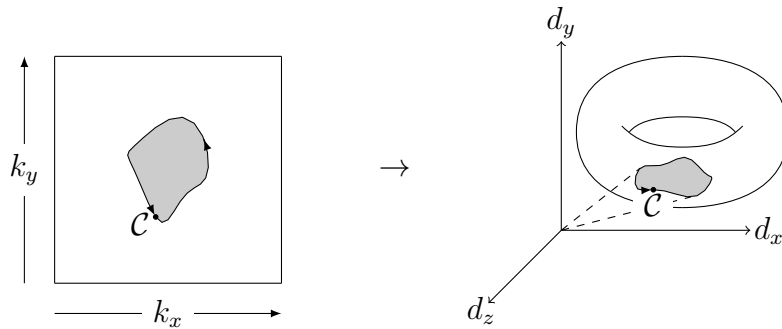
We know $\mathbf{k}$ lives on the surface of a torus, since both k_x and k_y are periodic with some maximum value k_x^{Max} and k_y^{Max} . Furthermore, $\mathbf{d}$ (and by extension $\hat{H}$) live in the space $\mathbb{R}^3$. Thus, the mapping $\mathbf{d}$ is

$$\mathbf{d} : S^1 \times S^1 \rightarrow \mathbb{R}^3 \quad (2.1.46)$$

$$\mathbf{d} : (k_x, k_y) \mapsto \mathbf{d}(\mathbf{k}) \quad (2.1.47)$$

The map is embedding a two-dimensional surface (the torus) into three-dimensional flat space ($\mathbb{R}^3$). There are no limits on how you can do this embedding, the torus can be as twisted and crinkled up as you like so long as the embedding respects the topology of the Brillouin zone (it can't include discontinuities or cuts or other abusive things).

When we want to calculate the Berry phase around a loop in $\mathbf{k}$ -space, all we have to do is look at the corresponding loop in $\mathbf{d}$ -space. Then we integrate the Berry connection over some surface that has this loop as its boundary. The Berry connection actually looks like a radiating monopole so we can simplify further, knowing that we just need to find the solid angle subtended by the curve around the origin and divide it by two.



What about the Chern number? To calculate this all we need to do is integrate the Berry curvature across the whole surface of the torus in $\mathbf{d}$ -space. If the embedding of the torus does

not contain the origin, then the value of this integral is necessarily zero. This is because the torus has an inside and an outside, everywhere that the surface is presented with the inside facing the origin we have a positive contribution to the integral, however when the outside is facing the origin we get a negative contribution. If the origin is outside the torus then we have exactly equal negative and positive contributions and everything cancels out!

Now suppose the torus is embedded such that it contains the origin, then we will have a surface that catches all of the Berry flux on its inside² and the solid angle subtended is 4π ! This means that the Chern number will be +1. Furthermore, you could imagine wrapping the torus around the origin twice, so that it effectively contains two origins, now the Chern number will be +2. You could turn it inside out and then wrap it around the origin too, that would give $Q = -1$ and so on...

2.1.4 Final Notes

We close this chapter with an explanation of the consequences of Chern number in topological insulators.

Smoothly Deforming the Hamiltonian

Now suppose that we have a pair of quantum systems that live in the same Hilbert space, described by $\hat{H}_1$ and $\hat{H}_2$. Suppose both of these Hamiltonians represent insulating materials. This means that the bands in these materials are separated across all of the Brillouin zone by a band gap. We might want to ask the question: *Is it possible to smoothly deform $\hat{H}_1$ into $\hat{H}_2$ without ever crossing a point where the bands touch and the material becomes conductive?*

The answer to this is that it is only possible if the bands of $\hat{H}_1$ and $\hat{H}_2$ have the same Chern numbers. This is because the Chern number is always restricted to have integer value, it cannot continuously change as we smoothly deform the Hamiltonian. In order to change the Chern number of a band we must somewhere cross a point where it is impossible to define a Chern number at all, allowing it to jump discontinuously when we leave that point. This is precisely what happens when two bands touch, the description of a single band becomes ill-defined and our calculation for Chern number stops working.

We can illustrate this with the example in §2.1.3. Smoothly deforming our Hamiltonian amounts to changing the shape of the torus embedded in $\mathbf{d}$ -space. If any point on the surface of the torus touches the origin, we have a pair of zero-energy states appearing in our system, this means the bands touch and the system is conductive. If the band has a Chern number of one, this means that the torus contains the origin. If we wanted to smoothly deform this to have zero Chern number, we would need to move the torus such that the origin was now outside. At some point in this deformation the torus surface will have to touch the origin, creating a conducting Hamiltonian at that point.

Thus we can classify all Hamiltonians according to the Chern number of their bands. It is always impossible to smoothly deform a Hamiltonian into one with a different Chern number while remaining insulating.

²Maybe you spotted this is not quite true, the torus is not a sphere so there will be some bits where there are outside faces presented to the origin, such as when you hit the hole in the doughnut! These will be cancelled out by the additional inside faces and so overall you still get 4π .

This allows us to make a rough justification for why conducting edge states must appear wherever we have a material with nonzero Chern number. Suppose we have a large material where the Hamiltonian is allowed to vary slowly over space, defined by $\hat{H}(\mathbf{r})$. Now let's say that in some region this Hamiltonian looks locally like it has non-zero Chern number, $\hat{H}_{\text{topological}}$, but in a different, distant region it looks locally like it has zero Chern number, $\hat{H}_{\text{trivial}}$. We expect that between these regions the Hamiltonian must be gradually transforming between $\hat{H}_{\text{topological}}$ and $\hat{H}_{\text{trivial}}$. As we have described, it is impossible for our Hamiltonian to make this transformation without creating a conducting surface somewhere between these two regions – edge states must appear! Loosely speaking, we can view the vacuum as a zero Chern number material, which gives us an intuitive picture for why we might expect materials with non-zero Chern number to always have edge states appearing on their surface, where the Hamiltonian has to transition to the vacuum.

Chern Number and Globally Smooth Gauges

If you thought carefully about the gauge that we chose in equation 2.1.36 you might notice that it has a problem, There is actually a discontinuity at $\phi = \pi$ where $e^{i\phi}$ suddenly jumps as you cross over to $\phi = -\pi$. This is no oversight, it turns out that it is actually completely impossible to define a globally smooth set of states $|\psi(\mathbf{d})\rangle$ for any band that has nonzero Chern number!

This is because if $|\psi(\mathbf{d})\rangle$ were continuous everywhere, then the Berry connection would also have to be continuous everywhere. This would mean that we could invoke Stokes' theorem when calculating the Chern number,

$$\mathcal{Q} = \frac{1}{2\pi} \int_{\mathcal{P}} B \quad (2.1.48)$$

$$= \frac{1}{2\pi} \int_{\partial\mathcal{P}} A. \quad (2.1.49)$$

The problem is that the Berry curvature integral is over the whole space $\mathcal{P}$. Since $\mathcal{P}$ is a closed manifold, it must have no boundary, so eqn. 2.1.49 must be equal to zero. Another way of understanding this is to realise that if you were to discretise this integral into a lattice of states, every edge would be an internal edge and the total would always have to be zero!

In this sense, the Chern number can be regarded as something like a measure of the amount of discontinuity that *must* be present regardless of the gauge you choose for the space.

2.2 The QWZ Model and its Amorphous Counterpart

For the rest of this document, we will use the Qi-Wu-Zhang (QWZ) model as our example for a topological insulator, first introduced in [?]. This is because the QWZ model is a simple toy model for a two-dimensional topological insulator with nearest-neighbour interactions on a square lattice.

We work in a similar Hilbert space as that used in §??, on a lattice of unit cells indexed by a two dimensional lattice vector $\mathbf{m}$, where each cell contains two sites – the QWZ model has two bands. Thus the Hilbert space of the system is spanned by states of the form

$$|\psi\rangle = |m_x\rangle \otimes |m_y\rangle \otimes |\alpha\rangle. \quad (2.2.1)$$

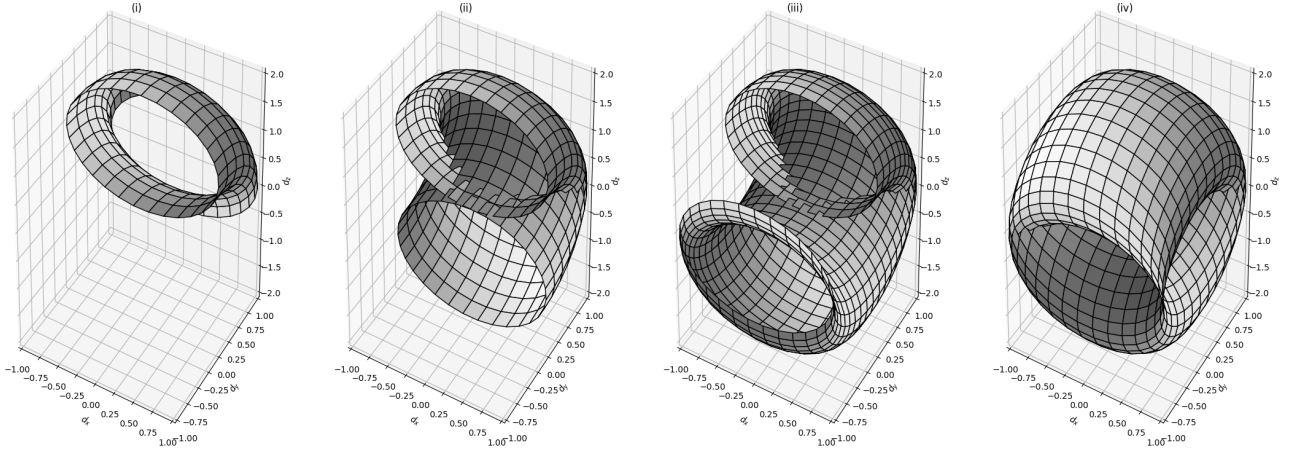


Figure 2.1: The embedding of the Brillouin zone in $\mathbf{d}$ -space for the QWZ model. This is shown in four progressive cutaways to ensure the internal structure can be seen. In all four images k_x spans the interval $(0, 2\pi]$, however in (i) k_y spans $(0, \frac{\pi}{2}]$, in (ii) k_y spans $(0, \pi]$, in (iii) k_y spans $(0, \frac{3\pi}{2}]$, and in (iv) k_y spans $(0, 2\pi]$.

The lattice has size (L_x, L_y) , so we have

$$m_x \in \{0, \dots, L_x - 1\} \quad (2.2.2)$$

$$m_y \in \{0, \dots, L_y - 1\} \quad (2.2.3)$$

$$\alpha \in \{0, 1\}. \quad (2.2.4)$$

The QWZ model is defined by three terms, a hopping between unit cells in the x -direction, a hopping in the y -direction, and a staggered on-site potential whose strength is determined by a parameter u .

$$\begin{aligned} \hat{H} = & \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_x\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_y\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_y}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m}\rangle \langle \mathbf{m}| \otimes u \sigma_z, \end{aligned} \quad (2.2.5)$$

where $\mathbf{1}_x$ and $\mathbf{1}_y$ represent the unit displacement in their respective directions,

$$\mathbf{1}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{1}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (2.2.6)$$

Using Bloch's theorem, we can find the bulk Hamiltonian,

$$\hat{H}(\mathbf{k}) = \begin{pmatrix} \sin k_x & \\ \sin k_y & \\ u + \cos k_x + \cos k_y & \end{pmatrix} \cdot \boldsymbol{\sigma}. \quad (2.2.7)$$

Here, k_x and k_y determine our position in the Brillouin zone. As explained in §2.1.3, this represents the embedding of a torus in $\mathbf{d}$ -space, shown in fig. 2.1.

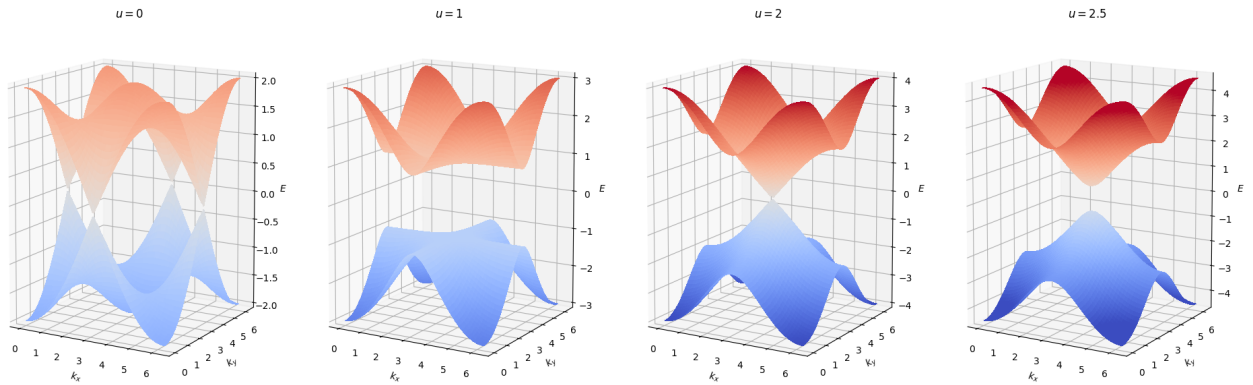


Figure 2.2: Four examples of the band structure of the QWZ model are shown, for varying u values between 0 and 2.5. At $u = 0$ and 2, the bands touch and Chern number is undefined. At $u = 1$, $Q = +1$, and at $u = 2.5$, $Q = 0$.

The u -parameter has the effect of translating the torus by a fixed amount in the z -direction. From this we can see that the QWZ model only has non-zero Chern number for values of u between -2 and 2. This is because translating the torus by any more than this will place the origin completely outside. The Chern number for the QWZ model is

$$Q = \begin{cases} -1 & : -2 < u < 0 \\ 1 & : 0 < u < 2 \\ 0 & : \text{otherwise} \end{cases}, \quad (2.2.8)$$

not including the points at $u = -2, 0$ and 2 , at which the Chern number is undefined because the band gap closes and the material is conducting. Some example band structures for varying values of u are shown in fig. 2.2.

2.2.1 Strip Geometry

In much of the following discussion, we will be studying the QWZ model in a system with strip geometry. This means that the lattice is periodic and uniform in the y -direction, but is allowed to be non-uniform or have open boundaries in the x -direction. We will allow u to vary in the x -direction, so add a subscript to indicate position-dependence $u \rightarrow u_{m_x}$. We can apply Bloch's theorem in the y -direction only, and our eigenstates take the following form,

$$|\psi\rangle = |k_y\rangle \otimes |v_{k_y}^{n,l}\rangle, \quad (2.2.9)$$

where $|k_y\rangle$ is a plane wave solution

$$|k_y\rangle = \frac{1}{\sqrt{L_y}} \sum_{m_y} e^{im_y k_y} |m_y\rangle, \quad (2.2.10)$$

and $|v_{k_y}^{n,l}\rangle$ is a $2L_x$ -element vector in the remaining part of the Hilbert space $\mathcal{H}_x \otimes \mathcal{H}_{\text{int}}$. The n index labels which band the state is in, and the l index differentiates between the states in each

band.

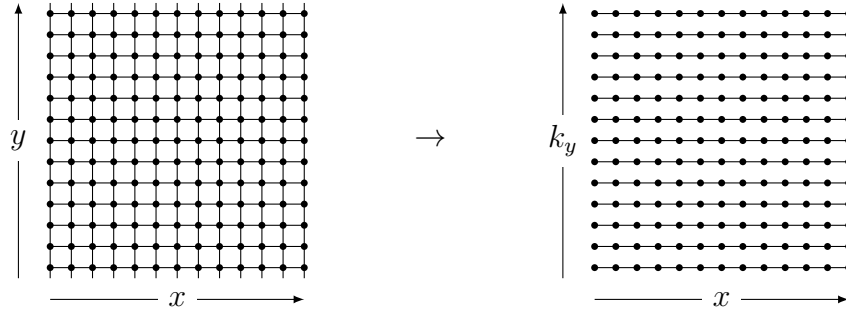
$$n \in \{0, 1\} \quad (2.2.11)$$

$$l \in \{0, \dots, L_x - 1\}. \quad (2.2.12)$$

The v -states are solutions of the semi-bulk Hamiltonian

$$\begin{aligned} \hat{H}(k_y) = & \sum_{m_x} |m_x + 1\rangle \langle m_x| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{m_x} |m_x\rangle \langle m_x| \otimes [(u_{m_x} + \cos k_y) \sigma_z + \sin k_y \sigma_y]. \end{aligned} \quad (2.2.13)$$

We have effectively reduced our system from a lattice of interacting sites in x and y , to a set of chain models in x , that are each labelled by a value of k_y .



We can rewrite this, in terms of the 2×2 matrices A and B ,

$$A = (u + \cos k_y) \sigma_z + \sin k_y \sigma_y, \quad (2.2.14)$$

$$B = \frac{\sigma_z + i\sigma_x}{2}, \quad (2.2.15)$$

and express our wavevectors in terms of a set of L_x two-element states ψ_n

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{L_x-1} \end{pmatrix}, \quad (2.2.16)$$

to arrive at a set of equations for the eigenvalues of the semi-bulk Hamiltonian.

$$\text{Bulk: } A\psi_n + B\psi_{n+1} + B^\dagger\psi_{n-1} = \lambda\psi_n \quad (2.2.17)$$

$$\text{Left Edge: } A\psi_1 + B\psi_2 = \lambda\psi_1 \quad (2.2.18)$$

$$\text{Right Edge: } A\psi_{L_x} + B^\dagger\psi_{L_x-1} = \lambda\psi_{L_x} \quad (2.2.19)$$

The band structure for this system in both periodic boundaries and strip geometry is shown in fig. 2.3. We can now look at solving this analytically for the bulk and edge states. We will omit this calculation and print only the form of the results, however a full derivation can be found in appendix ???. Note that edge states are only present for certain values of k_y , which can be seen in figure 2.3, where the edge states only separate from the bulk in a region around $k_y = \pi$.

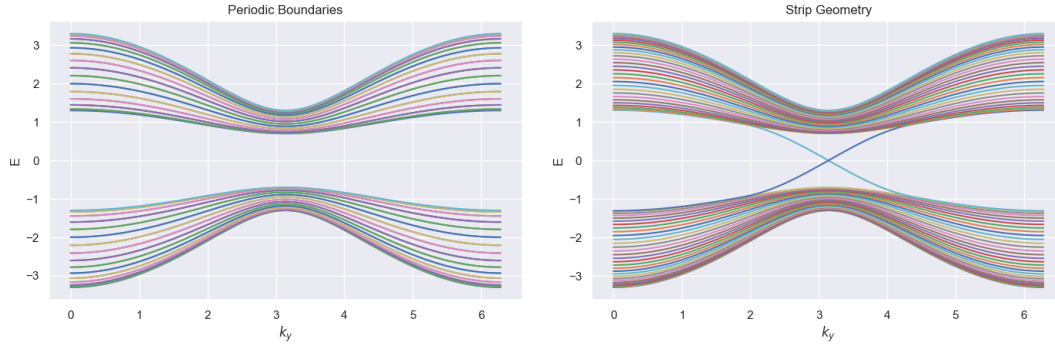


Figure 2.3: An example band structure for a QWZ model with $u = 1.3$ and $L_x = 30$. On the left is the example for periodic boundary conditions, showing two well-separated bands. On the right is the case for open boundary conditions. Two edge states are shown leaving the bulk and crossing the gap, with the descending state being a left edge state, and the ascending state being a right edge state.

Edge states have the form

$$|\psi_{k_y}^{\text{left}}\rangle = \sqrt{1 - \xi_l^2} \sum_{m=1}^{L_x} \xi_l^{m-1} |m\rangle \otimes \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (2.2.20)$$

where

$$\xi_l = -u - \cos k_y. \quad (2.2.21)$$

This state has energy

$$E_l = -\sin k_y. \quad (2.2.22)$$

Similarly, the right edge state is given by

$$|\psi_{k_y}^{\text{right}}\rangle = \sqrt{1 - \xi_r^2} \sum_{m=1}^{L_x} \xi_r^{L_x-m} |m\rangle \otimes \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (2.2.23)$$

with the extra conditions

$$\xi_r = -u - \cos k \quad (2.2.24)$$

$$E_r = \sin k. \quad (2.2.25)$$

These edge states are only valid solutions for k_y in the region

$$-1 < u + \cos k_y < 1. \quad (2.2.26)$$

These limits can be seen in fig. 2.3, where the edge states peel away from the bands to cross the energy gap.

The bulk states are less straightforward to obtain exact solutions for, we give the general form

for them and leave the details to appendix ??.

$$|\psi_{k_y}^\pm\rangle = \frac{1}{\sqrt{L_x}} \sum |m\rangle \otimes [\alpha e^{i\theta m} |\phi_\theta^\pm\rangle + \beta e^{-i\theta m} |\phi_\theta^{\pm*}\rangle], \quad (2.2.27)$$

where the exact values of θ , α , β and $|\phi_\theta^\pm\rangle$ can be found by solving some fairly unpleasant equations.

Here we mention a final caveat, These calculations are only strictly valid in the limit of a large system. This is because we have made the assumption the edge states can be calculated by taking into consideration only one edge and ignoring the effect of the other. The edge states decay exponentially through the system, so we are relying on the assumption that the edge state has effectively decayed to nothing before reaching the other side. When this condition is relaxed the edge states are able to couple to one another. This has a two effects, the first being that at the crossing of edge states, our energy eigenvalues become positive and negative superposition states of both edge states. The other is that the left state coupling to the right state shifts their energy levels such that no states cross $E = 0$ – there is no longer a crossing of energy levels for the left and right edge states.

2.2.2 The Amorphous QWZ Model

2.3 Local Markers

2.3.1 Kitaev's Local Marker

2.3.2 The Bott Index

2.3.3 The Chern Marker

Bibliography

- [1] M. Berry. Quantal phase factors accompanying adiabatic changes. *Proc. R. Soc. Lond. A*, 392(1802):45–57, March 1984.
- [2] J. K. Asbóth, L. Oroszlány, and A. Pályi. *A Short Course on Topological Insulators: Band Structure and Edge States in One and Two Dimensions*. Lecture Notes in Physics. Springer International Publishing, 2016.
- [3] R. Resta. Geometry and Topology in Electronic Structure Theory. Lecture Notes, 2020.
- [4] A. Bohm, A. Mostafazadeh, H. Koizumi, Q. Niu, and J. Zwanziger. *The Geometric Phase in Quantum Systems: Foundations, Mathematical Concepts, and Applications in Molecular and Condensed Matter Physics*. Theoretical and Mathematical Physics. Springer-Verlag, Berlin Heidelberg, 2003.
- [5] D. Vanderbilt. *Berry phases in electronic structure theory*. Cambridge University Press, 2018. OCLC: 1081085754.

Chapter 3

The Crosshair Marker

Chapter 4

Kitaev's Honeycomb Model

Chapter 5

The Amorphous Kitaev Model

Chapter 6

The Star Lattice Kitaev-Heisenberg Model

Chapter 7

Conclusion and Outlook